Practice workbook · Spot selection and iteration, and pick the right loop

Naming the construct shown in a flowchart or keyword, identifying the kind of loop in an algorithm, and explaining why a condition-controlled loop suits a task better than a count-controlled one.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-06

1. Question 6 · part b.iv · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.iv only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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1. Question 6 · part b.iv · 2 marks

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

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1. Question 6 · part b.iv · 2 marks

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Worked-example retry · Attempt part b.iv only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

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(iii) State the purpose of the algorithm.

(1)

.....

.....

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

.....

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(Total for Question 6 = 13 marks)

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1. Question 6 · part b.iv · 2 marks · Working and answer

Working and answer

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2. Question 1 · part c · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part c only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

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2. Question 1 · part c · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part c only.

(c) **Figure 1** shows a programming construct in a flowchart.

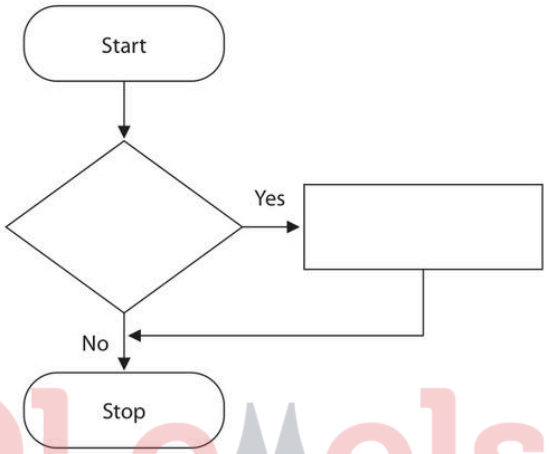


Figure 1

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

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2. Question 1 · part c · 1 mark · Working and answer

Working and answer

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3. Question 6 · part a · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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P 7 5 7 3 6 A 0 2 2 2 8

3. Question 6 · part a · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a only.

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(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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3. Question 6 · part a · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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4. Question 1 · part a · 1 mark · Working and answer

Working and answer

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